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sklearn documentation.

Logistic Regression Hyper parameter.

watch sklearn documentation

Naive Bayes Classifier.

Conditional Probability.

A, B

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

given, $P(B) \neq 0$

Roll 2 dice ~~two times.~~ together, D_1

D_2	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

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1) $P(A = 5) = \frac{1}{36}$.

2) $x_1 + x_2 <= 10$:

$$P(B) = P(x_1 + x_2 <= 10) = \frac{33}{36}$$

3) $\underbrace{x_1 = 5}_{A}$ given $\underbrace{x_1 + x_2 <= 10}_{B}$.

$P(x_1 = 5 \mid x_1 + x_2 <= 10)$

$$P(A|B) = \frac{5}{33}$$

using table.

using formula: $P(A|B) = \frac{P(A \cap B)}{P(B)}$

$A = \{x_1 = 5\}$, $B = \{x_1 + x_2 <= 10\}$

$P(A \cap B) = \frac{5}{36}$ $\left\{\begin{matrix} 5+1 \\ 5+2 \\ 5+3 \\ 5+4 \end{matrix}\right\}$

$P(B) = \frac{33}{36}$.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{5/36}{33/36} = \frac{5}{33}$$

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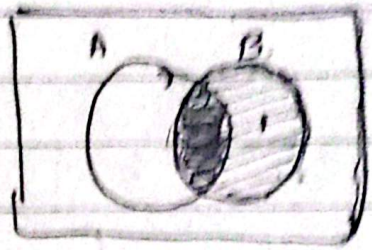
Independent Events

$$P(A \cap B) = P(A) \cdot P(B)$$

~~eg: tossing, baby gender~~

$$\boxed{P(A|B) = \frac{n(A \cap B)}{n(B)}} \quad \text{--- (1)}$$



From above:

$$P(A) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{n(A \cap B)}{n(S)}}{\frac{n(B)}{n(S)}}$$

$$\boxed{P(A) = \frac{n(A \cap B)}{n(B)}} \quad \text{--- (2)}$$

$$\therefore P(A|B) = P(A)$$

Mutually Exclusive Events.

$$P(A \cap B) = 0$$

eg: tossing, baby gender.

$$P(A \cap B) = \frac{P(A \cap B)}{P(B)} = 0.$$

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Bayes Theorem:

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

likelihood → $P(B|A)$; prior → $P(A)$; evidence → $P(B)$; postieror → $P(A|B)$

(Probability of A given B)

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$\Rightarrow A \cap B = B \cap A.$

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

$$= \frac{P(A \cap B)}{P(A)}$$

$$\boxed{P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)}}$$

(a) Marker factory has three machines, ~~then~~ (M_1, M_2, M_3) the percentages of markers made by the factory:

$M_1 = 20\%$, $M_2 = 30\%$, $M_3 = 50\%$

Manufacture defect: 5%, 3%, 1%

If we choose the marker from the bunch of markes, it comes out to be a defective piece. What is it's probability to be from M_3.

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$P(M_1) = \frac{20}{100} = \frac{1}{5}$, $P(M_2) = \frac{30}{100}$, $P(M_3) = \frac{50}{100} = \frac{1}{2}$

$P(D|M_1) = \frac{5}{100} = \frac{1}{20}$, $P(D|M_2) = \frac{3}{100}$, $P(D|M_3) = \frac{1}{100}$

$P(M_3|D) = ?$

$$P(M_3|D) = \frac{P(D|M_3) \cdot P(M_3)}{P(D)}$$

$$= \frac{\frac{1}{100} \cdot \frac{1}{2}}{\frac{1}{20} + \frac{3}{100} + \frac{1}{100}}$$

$$= \frac{1}{\cancel{10}}$$

$$P(D) = \frac{3}{125}$$

$$\therefore \quad P(M_3|D) = \frac{\cancel{P(D|M_3)} \ \frac{1}{100} \cdot \frac{1}{2}}{3/125}$$

$$= \frac{5}{24}.$$

OR

$$P(M_3|D) = \frac{P(M_3 \cap D)}{P(D)} \rightarrow \frac{0.5 \times 50}{100}$$

$$= \frac{1/200}{3/125}$$

$$= \frac{5}{24}.$$